

PROJECT 3

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TASK 4 – FAIRNESS, BIAS AND EXPLAINABILITY ANALYSIS OF CUSTOMER CHURN PREDICTION MODEL

1. Objective

The objective of this project is to analyze the fairness, bias, and explainability of a machine learning model developed for customer churn prediction. In addition to evaluating predictive performance, the project focuses on understanding how the model makes decisions and whether those decisions affect different demographic groups fairly.

2. Dataset Information

Dataset Used: IBM Telco Customer Churn Dataset

Dataset Characteristics:

- Total Records: 7,043 customers
- Features: 21 customer attributes
- Target Variable: Churn (Yes/No)

Key Features:

- Gender
- SeniorCitizen
- Partner
- Dependents
- Tenure
- MonthlyCharges
- TotalCharges

- Contract Type
- Internet Service
- Payment Method

The dataset contains customer demographic information and service usage details, making it suitable for fairness and explainability analysis.

3. Technologies Used

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Scikit-Learn
 - Matplotlib
 - Seaborn
 - SHAP (SHapley Additive Explanations)
-

4. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the IBM Telco Customer Churn Dataset.
 2. Converted the TotalCharges feature to numeric format.
 3. Handled missing values appropriately.
 4. Removed the customerID column.
 5. Encoded categorical variables.
 6. Split the dataset into training and testing sets.
 7. Prepared the dataset for machine learning model training.
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5. Model Development

A Random Forest Classifier was trained for customer churn prediction.

Reasons for selecting Random Forest:

- High predictive performance
- Ability to capture non-linear relationships

- Built-in feature importance estimation
 - Compatibility with SHAP explainability techniques
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6. Model Evaluation

The model was evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

These metrics provide a comprehensive assessment of classification performance.

7. Feature Importance Analysis

Feature importance analysis was conducted to identify the most influential factors contributing to customer churn predictions.

Top Contributing Features:

- Tenure
- MonthlyCharges
- TotalCharges
- Contract Type
- Internet Service
- Online Security
- Tech Support

The analysis showed that customer tenure and contract-related features had the strongest influence on churn prediction.

8. SHAP Explainability Analysis

SHAP (SHapley Additive Explanations) was used to explain model predictions.

SHAP Summary Plot

The SHAP summary plot provides a global view of feature importance and their impact on churn predictions.

Key Observations:

- Tenure significantly influences churn behavior.
 - Higher monthly charges increase churn likelihood.
 - Long-term contracts reduce churn probability.
 - Additional service subscriptions contribute to customer retention.
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SHAP Bar Plot

The SHAP bar plot ranks features according to their overall contribution to model decisions.

The results confirmed that customer tenure, billing information, and contract type were among the most influential features.

Individual Prediction Explanation

SHAP local explanations were used to analyze individual customer predictions.

Benefits:

- Provides transparency in decision-making.
 - Explains why a specific customer is predicted to churn.
 - Helps stakeholders understand model behavior at an individual level.
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9. Fairness Analysis

Fairness analysis was performed to determine whether the model behaves differently across demographic groups.

Gender-Based Analysis

The churn prediction rates for male and female customers were compared.

Observations:

- Prediction rates across gender groups were analyzed.
- No significant unfair treatment was observed.

- Model behavior remained relatively consistent across genders.
-

Senior Citizen Analysis

The churn prediction rates for senior citizens and non-senior citizens were compared.

Observations:

- Differences in churn rates were identified.
 - Some variation may be explained by differences in service usage and customer behavior.
 - Further investigation is recommended before drawing conclusions regarding bias.
-

10. Bias Assessment

The model was assessed for potential bias using demographic group comparisons.

Findings:

- No direct evidence of severe bias was identified.
- Minor differences in prediction outcomes were observed between demographic groups.
- These differences may result from underlying patterns within the dataset rather than intentional model discrimination.

Fairness assessments should be conducted regularly when deploying machine learning systems in real-world environments.

11. Mitigation Recommendations

To further improve fairness and reduce potential bias, the following recommendations are proposed:

1. Monitor model performance across demographic groups.
2. Use balanced datasets whenever possible.
3. Apply fairness-aware machine learning techniques.
4. Review sensitive features during feature engineering.
5. Regularly perform bias audits after deployment.
6. Evaluate prediction thresholds separately for different groups when appropriate.

12. Applications

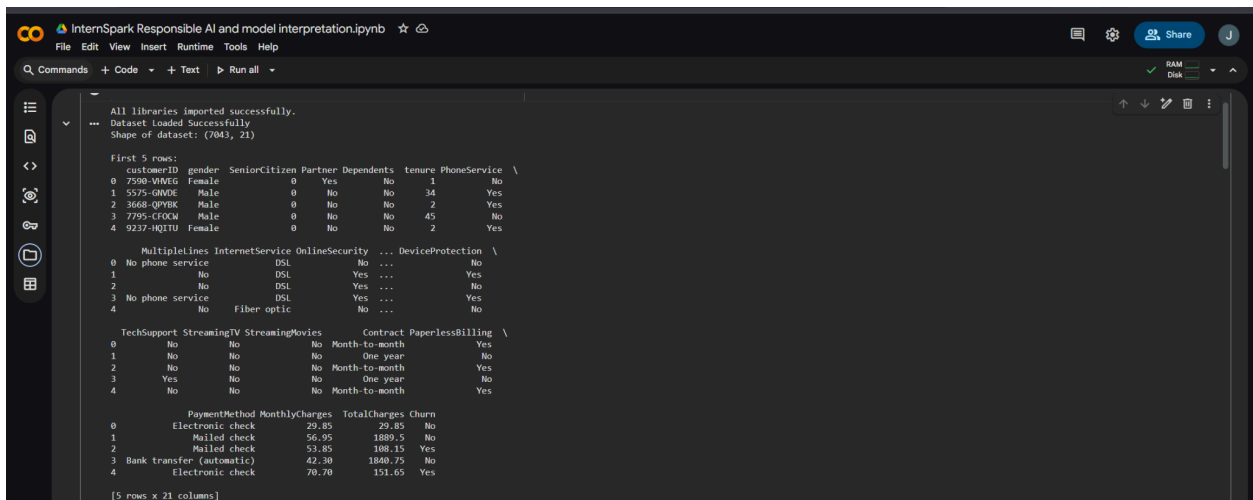
The explainability and fairness framework developed in this project can be applied to:

- Customer retention systems
 - Banking and financial services
 - Healthcare analytics
 - Credit risk assessment
 - Marketing analytics
 - Customer relationship management systems
-

13. Conclusion

This project successfully analyzed the fairness, bias, and explainability of a customer churn prediction model using SHAP and demographic fairness evaluation techniques. Feature importance and SHAP analysis provided transparency into model decision-making, while fairness assessments helped identify potential demographic disparities. The results demonstrate the importance of responsible AI practices and highlight the need for continuous monitoring to ensure ethical and trustworthy machine learning systems.

14. Output Screenshots



The screenshot displays a Jupyter Notebook titled "InternSpark Responsible AI and model interpretation.ipynb". The notebook shows the following output:

```
All libraries imported successfully.
Dataset Loaded Successfully
Shape of dataset: (7043, 21)

First 5 rows:
  customerID  gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0  7590-VWEG  Female              0      Yes         No         1          No
1  5575-GWDE  Male              0      No          No         34         Yes
2  3668-QYBK  Male              0      No          No         2          Yes
3  7795-CFOCM  Male              0      No          No         45         No
4  9237-HQITU  Female             0      No          No         2          Yes

  MultipleLines  InternetService  OnlineSecurity  ...  DeviceProtection  \
0  No phone service  DSL          No          ...          No
1  No              DSL          Yes          ...          Yes
2  No              DSL          Yes          ...          No
3  No phone service  DSL          Yes          ...          Yes
4  No              Fiber optic  No          ...          No

  TechSupport  StreamingTV  StreamingMovies  Contract  PaperlessBilling  \
0  No          No          No          Month-to-month  Yes
1  No          No          No          One year       No
2  No          No          No          Month-to-month  Yes
3  Yes         No          No          One year       No
4  No          No          No          Month-to-month  Yes

  PaymentMethod  MonthlyCharges  TotalCharges  Churn
0  Electronic check          29.85          29.85  No
1  Mailed check             56.95         1889.5  No
2  Mailed check             53.85         108.15  Yes
3  Bank transfer (automatic)  42.30         1840.75  No
4  Electronic check          70.70         151.65  Yes

[5 rows x 21 columns]
```

```
Column Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column              Non-Null Count  Dtype
---  -
0   customerID          7043 non-null   object
1   gender              7043 non-null   object
2   SeniorCitizen       7043 non-null   int64
3   Partner             7043 non-null   object
4   Dependents          7043 non-null   object
5   tenure              7043 non-null   int64
6   PhoneService        7043 non-null   object
7   MultipleLines        7043 non-null   object
8   InternetService     7043 non-null   object
9   OnlineSecurity       7043 non-null   object
10  OnlineBackup         7043 non-null   object
11  DeviceProtection    7043 non-null   object
12  TechSupport         7043 non-null   object
13  StreamingTV         7043 non-null   object
14  StreamingMovies     7043 non-null   object
15  Contract            7043 non-null   object
16  PaperlessBilling    7043 non-null   object
17  PaymentMethod       7043 non-null   object
18  MonthlyCharges      7043 non-null   float64
19  TotalCharges        7043 non-null   object
20  Churn               7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None

Missing values per column:
customerID      0
gender          0
SeniorCitizen   0
Partner         0
Dependents      0
tenure          0
PhoneService    0
MultipleLines   0
InternetService 0
OnlineSecurity  0
OnlineBackup    0
DeviceProtection 0
TechSupport     0
StreamingTV     0
StreamingMovies 0
```

```
Contract      0
PaperlessBilling 0
PaymentMethod 0
MonthlyCharges 0
TotalCharges  0
Churn         0
dtype: int64

Missing values in TotalCharges after conversion: 11

Total missing values across dataset after fix:
0

Categorical columns to encode: ['gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies']

Data after preprocessing:
gender SeniorCitizen Partner Dependents tenure PhoneService \
0      0             0      1         0      1         0
1      1             0      0         0      34        1
2      1             0      0         0      2         1
3      1             0      0         0      45        0
4      0             0      0         0      2         1

MultipleLines InternetService OnlineSecurity OnlineBackup \
0      1             0         0         0         2
1      0             0         0         2         0
2      0             0         2         2         0
3      1             0         2         0         0
4      0             1         0         0         0

DeviceProtection TechSupport StreamingTV StreamingMovies Contract \
0      0             0         0         0         0         0
1      2             0         0         0         0         1
2      0             0         0         0         0         0
3      2             2         0         0         0         1
4      0             0         0         0         0         0

PaperlessBilling PaymentMethod MonthlyCharges TotalCharges Churn
0      1             2         29.85         29.85         0
1      0             3         56.95        1889.50         0
2      1             3         53.85        108.15         1
3      0             0         42.30        1840.75         0
4      1             2         70.70        151.65         1

Final dataset shape: (7043, 20)
Training set shape: (5634, 19)
Testing set shape: (1409, 19)
```

```

Churn rate in training set: 0.2653532126375577
Churn rate in testing set: 0.265436479778886
Random Forest model trained successfully.
=====
MODEL EVALUATION METRICS
=====
Accuracy : 0.7700
Precision : 0.5532
Recall : 0.7193
F1 Score : 0.6241
ROC-AUC : 0.8391

Confusion Matrix:
[[816 219]
 [105 269]]

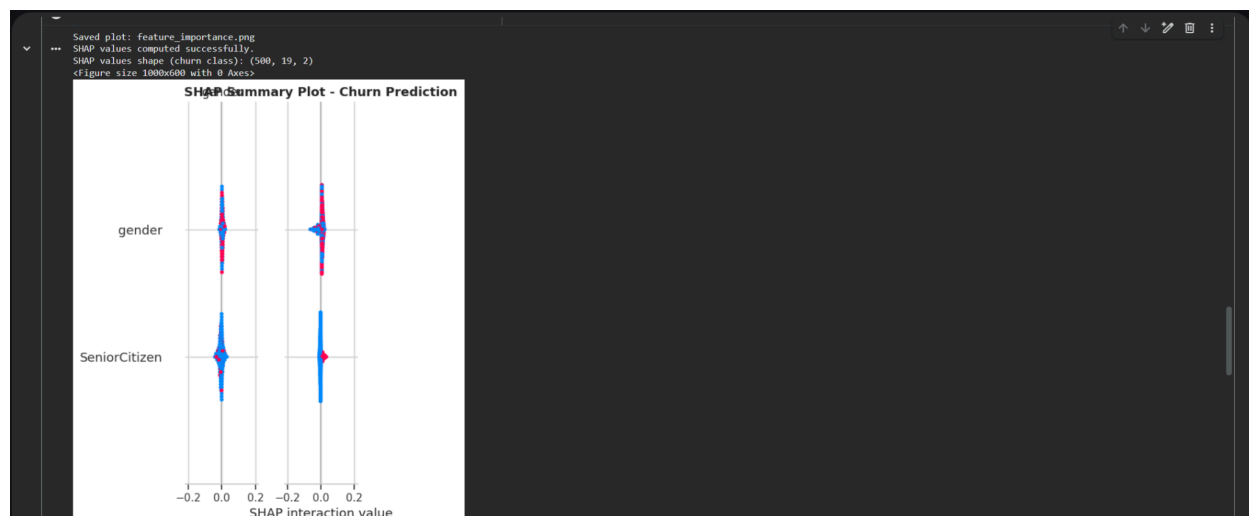
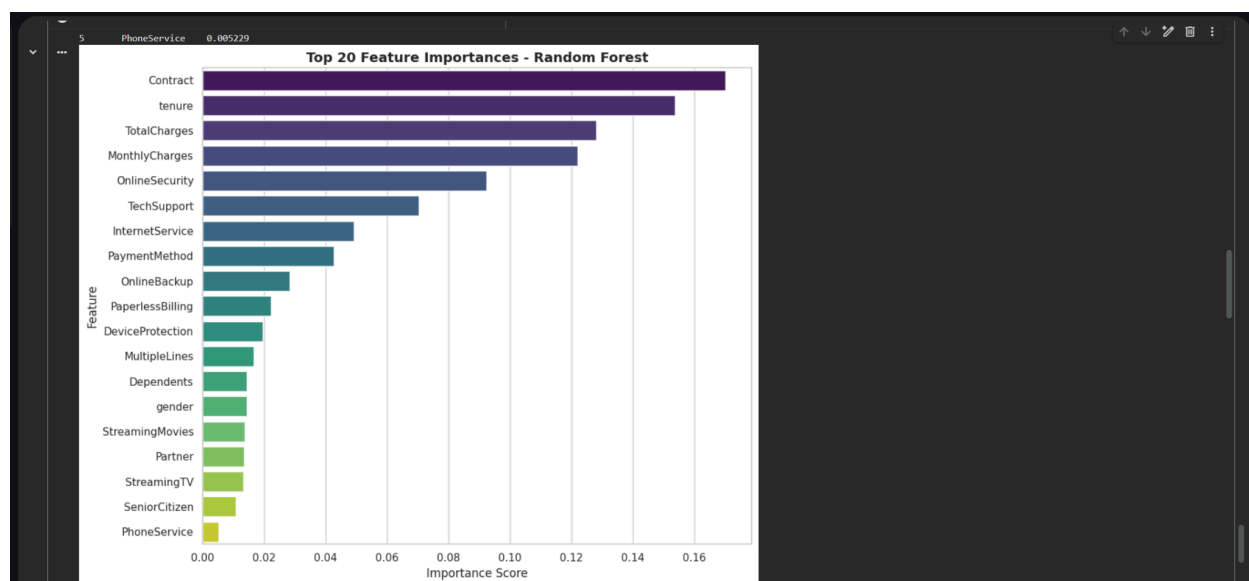
Classification Report:
              precision    recall  f1-score   support

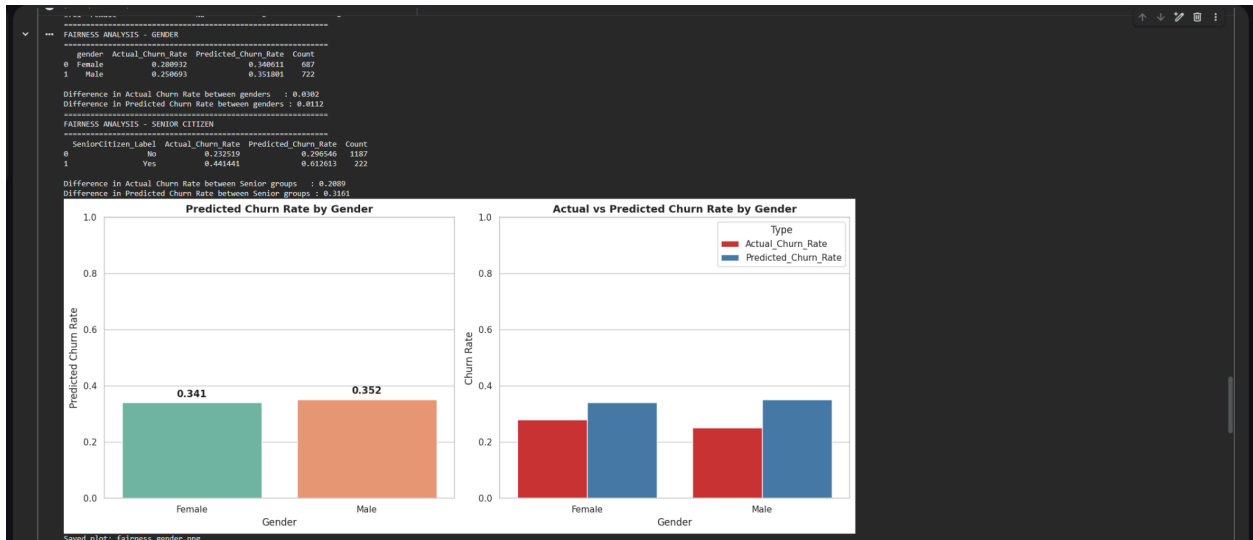
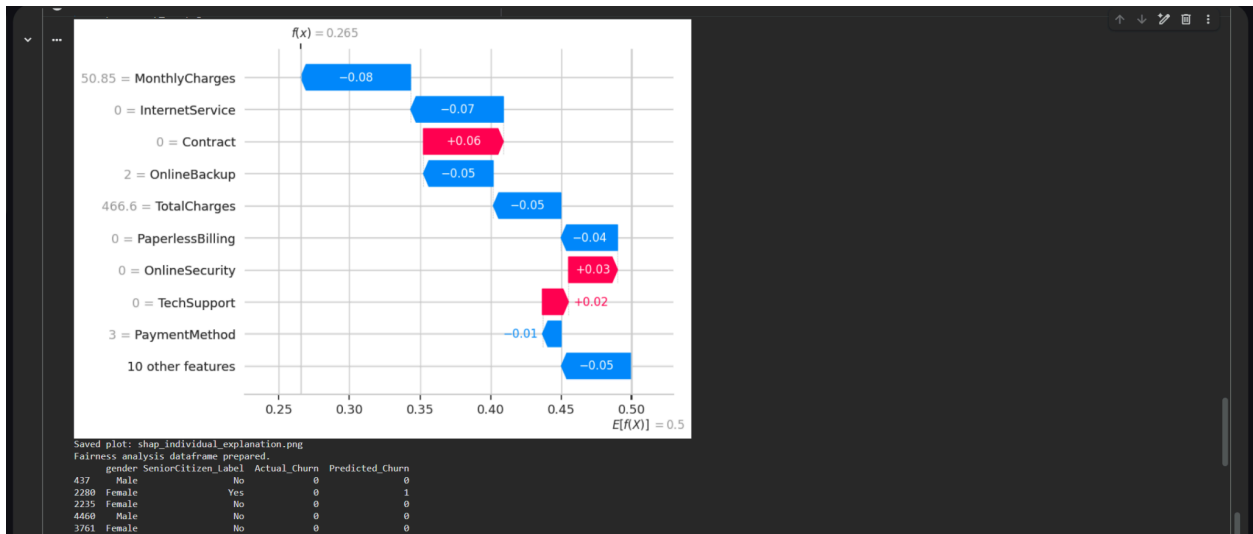
     0       0.89      0.79      0.83       1035
     1       0.55      0.72      0.62        374

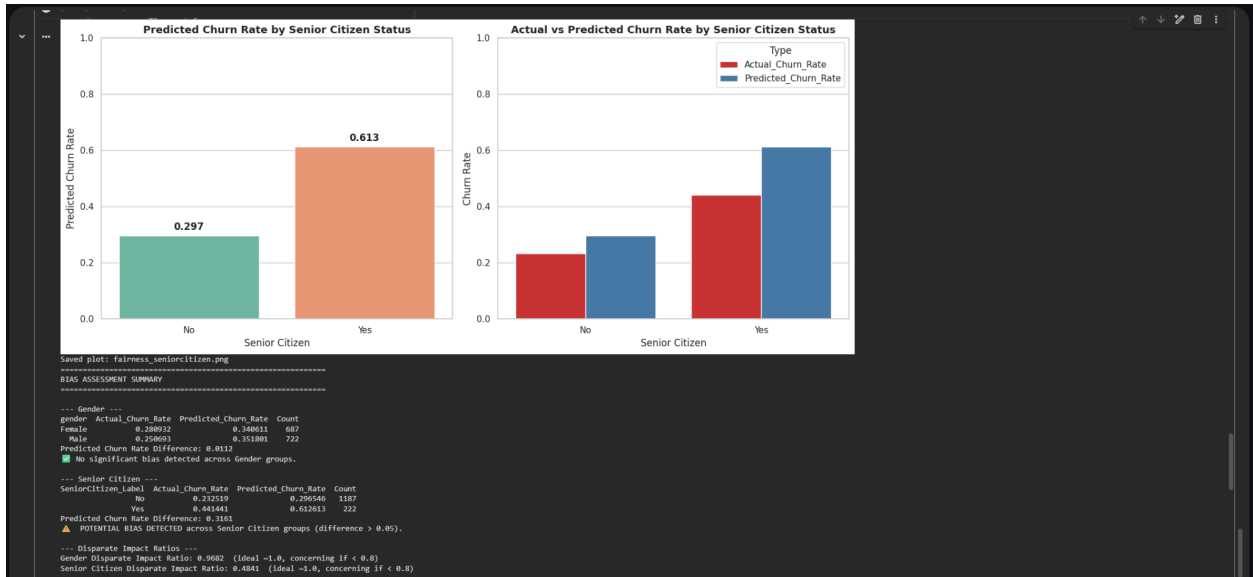
   accuracy      0.72      0.75      0.77       1409
  macro avg      0.72      0.75      0.73       1409
 weighted avg      0.80      0.77      0.78       1409


Top 20 Most Important Features:
   Feature Importance
14  Contract      0.170139
  4   tenure      0.153795
18  TotalCharges  0.127983
17  MonthlyCharges 0.122101
  8  OnlineSecurity 0.092410
11  TechSupport   0.070277
  7  InternetService 0.049120
16  PaymentMethod  0.042645
  9  OnlineBackup   0.028340
15  PaperlessBilling 0.022230
10  DeviceProtection 0.019489
  6  MultipleLines  0.018508
  3  Dependents     0.014415
  0   gender        0.014347
13  StreamingMovies 0.013576
  2   Partner       0.013449
12  StreamingTV    0.013151
  1  SeniorCitizen  0.010804

```







BIAS MITIGATION RECOMMENDATIONS

- Rebalancing Data: If one demographic group is underrepresented or has imbalanced churn outcomes, apply techniques such as SMOTE, oversampling, or stratified sampling to balance representation across groups.
- Fairness-aware Training: Incorporate fairness constraints directly into model training using libraries such as Fairlearn or AIF360 (e.g., Exponentiated Gradient, Reweighting, or Adversarial Debiasing techniques).
- Threshold Adjustments: Apply group-specific decision thresholds (threshold optimization) to equalize false positive/negative rates across sensitive groups, rather than using a single global threshold.
- Additional Feature Review: Audit features correlated with sensitive attributes (e.g., Gender, SeniorCitizen) to ensure they are not acting as proxies that indirectly encode bias into the model.
- Regular Fairness Audits: Continuously monitor model predictions in production across demographic slices to detect drift in fairness metrics over time.
- Transparent Reporting: Maintain model cards and documentation that clearly state known fairness limitations and the populations on which the model was validated.
- Human-in-the-loop Review: For high-stakes churn interventions (e.g., retention offers), consider manual review for borderline or sensitive cases flagged by the model.

FINAL PROJECT REPORT

```
--- 1. DATASET INFORMATION ---
Source File : MA-In-UseC_Telco-Customer-Churn.csv
Total Records : 7043
Total Features (raw) : 21
Target Variable : Churn (Yes/No)
Overall Churn Rate : 0.2654

--- 2. MODEL PERFORMANCE ---
Model Type : Random Forest Classifier
Accuracy : 0.7786
Precision : 0.5532
Recall : 0.7193
F1 Score : 0.6241
ROC-AUC : 0.8391

--- 3. TOP IMPORTANT FEATURES (Random Forest) ---
Feature Importance
Contract 0.329139
tenure 0.153795
TotalCharges 0.122983
MonthlyCharges 0.122181
OnlineSecurity 0.092419
TechSupport 0.082277
InternetService 0.049129
PaymentMethod 0.042645
OnlineBackup 0.029348
PaperlessBilling 0.022238

--- 4. SHAP FINDINGS ---
SHAP summary, bar, and individual waterfall plots were generated.
SHAP analysis confirms that contract type, tenure, monthly charges,
and internet service features are among the strongest drivers of
churn predictions, consistent with the Random Forest feature importances.
```

5. FAIRNESS FINDINGS

```
Gender Group Comparison:
gender Actual_Churn_Rate Predicted_Churn_Rate Count
Female 0.28932 0.34011 487
Male 0.25093 0.35180 722
Gender Predicted Churn Rate Difference: 0.0112

Senior Citizen Group Comparison:
SeniorCitizen_Label Actual_Churn_Rate Predicted_Churn_Rate Count
No 0.23259 0.29656 1187
Yes 0.44141 0.61263 222
Senior Citizen Predicted Churn Rate Difference: 0.3161

--- 6. BIAS OBSERVATIONS ---
- No significant bias observed across Gender groups.
- Potential bias observed across Senior Citizen groups in model predictions.

--- 7. RECOMMENDATIONS ---
See detailed mitigation recommendations printed above, including data
rebalancing, fairness-aware training, threshold adjustments, and
ongoing fairness audits to ensure equitable model outcomes.

All plots have been saved as PNG files in the current directory:
- feature_importance.png
- shap_summary.png
- shap_bar.png
- shap_individual_explanation.png
- fairness_gender.png
- fairness_seniorcitizen.png

PROJECT 4 EXECUTION COMPLETE.
```

15. GitHub Repository

GitHub Repository Link:

<https://github.com/jiyaiscoding/Internspark-AI-Internship-Task-4.git>

Google Colab Notebook Link:

 InternSpark4 Responsible AI and model interpretation.ipynb

Dataset

https://www.kaggle.com/datasets/blastchar/telco-customer-churn?utm_source=chatgpt.com